Dialogue System for Early Mental Illness Detection: Toward a Digital Twin Solution

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Agenda

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- Outcome of Literature Survey
- Problem Statement and Objectives
- Existing Methodology
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- Proposed enhancements/novelty
- Work done (Current status)
- Individual Contribution

Introduction (2-3 slides)

- **Global Context**: Mental health disorders have risen significantly, with depression affecting over 280 million people worldwide.
- **The Gap:** Barriers such as stigma, high costs, and insufficient clinical resources prevent many from seeking help.
- **The Solution:** Artificial Intelligence and "Digital Twin" frameworks can offer accessible, continuous assessment and personalized feedback.
- **Core Concept:** Unlike traditional systems that only detect the *presence* of depression, this project focuses on classifying the **severity** (No symptoms, Mild, Moderate/Severe) to prioritize critical cases

Literature Review (2-3 slides)

Author/year	Methodology	Advantages	Limitations
Hennemann et al. (2022)	**Ada Symptom Checker:** An app-based tool that uses AI to conduct interview-based communication, matching user symptoms against a medical database to predict conditions.	**High Usability:** Achieved a high System Usability Scale (SUS) score of 81.51% and is available in 7 languages across 10 million users. Validated on real patient data	**Diagnostic Accuracy:** Only 51% accuracy for the first condition suggestion. Participants still preferred face-to-face interviews over the app.
Fitzpatrick et al. (2017)	**Woebot:** A fully automated conversational agent delivering Cognitive Behavioral Therapy (CBT) to young adults with depression and anxiety symptoms	**Effective Intervention:** Significantly reduced depression symptoms (PHQ-9) and anxiety (GAD-7) in just two weeks. Users rated the bot high on empathy and therapeutic alliance	**Comparison Bias:** The control group only used an e-book, which may have inflated the perceived effectiveness of the chatbot. Results may not generalize to other populations.
Baker et al. (2020)	**Babylon Triage & Diagnostic System:** Uses probabilistic graphical models (Bayesian networks) to perform triage and diagnosis based on user evidence	**Explainability:** The use of Bayesian networks makes the decision-making process more interpretable and explainable compared to "black box" models. Comparable precision to human doctors.	**Scope:** The range of diseases it can diagnose is limited. It struggles with the complexity of diagnosing multiple concurrent conditions.
Philip et al. (2017)	**Embodied Conversational Agents (ECAs):** Virtual human agents that conduct diagnostic interviews for Major Depressive Disorders (MDD) based on DSM-5 criteria	**High Specificity:** Achieved a specificity of over 95% across three severity levels, minimizing false positives. High user acceptance and "AUC" of 0.71.	**Low Sensitivity:** Sensitivity for detecting mild symptoms was very low (21%). The trial population lacked diversity, limiting generalizability.

Literature Review (2-3 slides)

Author/year	Methodology	Advantages	Limitations
Denecke et al. (2020)	**SERMO:** A mobile app chatbot that uses NLP and lexicon-based techniques to monitor emotions and provide regulation recommendations.	**Emotion Tracking:** Offers comprehensive tools for emotion recognition, tracking, and regulation. Capable of working offline without internet access.	**Technical Limits:** The lexicon-based approach restricts the recognition of complex or unexpected user inputs. Low accuracy in detecting mild to moderate symptoms.
Bickmore et al. (2010)	**Relational Agent:** A virtual agent designed for patient education at hospital discharge to assist patients with depressive symptoms.	**Patient Preference:** Patients with depression rated the therapeutic relationship higher than non-depressed patients. Many preferred the agent over clinicians for psychoeducation	**Unknown Causality:** The direction of correlations between depressive symptoms and reported measures is unknown. Results might not apply to broader demographics
Semigran et al. (2015)	**Audit of Symptom Checkers:** Evaluated the diagnostic precision of 23 different symptom checkers using 46 emergency care vignettes.	**Comprehensive Benchmark:** Provided a broad evaluation of the state of the industry. Highlighted the gap between AI tools and human physicians.	**Low Precision:** Only one-third of the tools correctly predicted the diagnosis as the first option. Significantly outperformed by physicians (who achieved 84.3% vs. AI's 51.2% in top-3).
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Outcome of Literature Review

- **Limitation 1 (Binary Focus):** Most existing literature focuses on binary classification (Depressed vs. Not Depressed), missing the nuance of severity levels required for clinical prioritization.
- **Limitation 2 (Data Scarcity):** High-quality clinical datasets like E-DAIC are restricted, limiting reproducibility for general research.
- **Limitation 3 (Contextual Gaps):** Clinical interview models often struggle with informal, noisy text found in real-world social media interactions.
- **Opportunity:** There is a need to adapt severity classification models to handle accessible, informal text data (like Reddit) using robust modern architectures.

Problem Statement

- **Problem Statement:** "To develop and evaluate an NLP-based classification system capable of detecting the **severity** of mental health symptoms (No Symptoms, Mild, Moderate/Severe) from unstructured social media text, addressing the limitations of binary detection and data scarcity."

1. Replicate & Adapt: Adapt the multi-class severity logic (No symptoms, Mild, Moderate/Severe) from the base paper to a publicly available Reddit dataset.
2. Develop Parallel Models: Implement two distinct approaches:

Approach A: A "Paper-Faithful" BERT model using contextual augmentation.

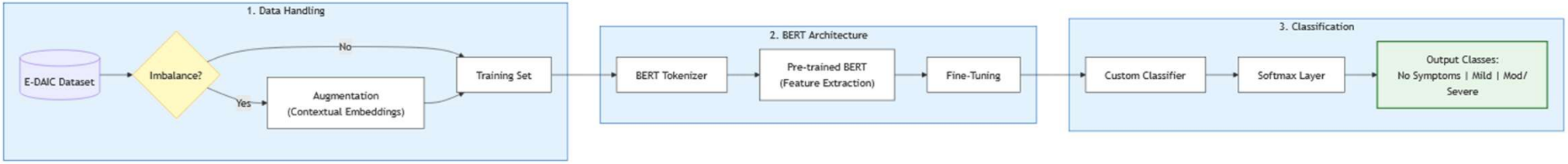
Research Objectives

1. Replicate & Adapt: Adapt the multi-class severity logic (No symptoms, Mild, Moderate/Severe) from the base paper to a publicly available Reddit dataset.
2. Develop Parallel Models: Implement two distinct approaches:

- Approach A: A "Paper-Faithful" BERT model using contextual augmentation.
- Approach B: A "Robust" RoBERTa model using class-weighted loss for informal text.

3. Comparative Evaluation: Compare the performance of both approaches using F1-score and Accuracy to determine the best method for analyzing noisy, user-generated mental health content

Existing Methodology



Dataset Details / Statistics

- **Dataset Name:** Mental Health Text Classification Dataset (Aggregated from Reddit).
- **Source:** r/depression, r/SuicideWatch, r/Anxiety, r/casualconversation.
- **Size:** ~50,000 text samples (approx).
- **Label Mapping (Adapting to Base Paper):**

1. *Original:* Normal → **Mapped:** No Symptoms (Class 0).
2. *Original:* Anxiety / Depression → **Mapped:** Mild/Moderate (Class 1).
3. *Original:* Suicidal → **Mapped:** Severe (Class 2).

- **Justification:** Provides a large-scale, accessible alternative to the restricted E-DAIC dataset while preserving the hierarchy of severity needed for the research.

Proposed enhancements/novelty

- **Approach A: (paper replication)**
  - **Model: BERT(bert-base-uncased)**
  - **Imbalance:** We generate synthetic samples for the "Severe" class using BERT embeddings to balance the dataset, mimicking the base paper's strategy.
- **Approach B (Proposed enhancement)**
  - **Model: RoBERTa(Roberta-base)**
  - **Novelty:** RoBERTa is optimized for informal internet text (like Reddit).
  - **Imbalance Handling: Class Weighted Loss**. Instead of generating synthetic data, we modify the CrossEntropyLoss function to penalize errors on the minority class ("Severe") more heavily.

Work done (Current status of work)

- We have completed reading the paper, finding out the relevant technologies and identified a suitable preprocessed dataset.
- Successfully mapped original 4 labels to the target 3 severity classes of mental health identification.
- Pranav Vinodh and Sanchit: Have started working on Approach A for the paper replication and to identify how well the existing system performs to enable a comparison study with approach B.
- Pranav Bhat and Rijul Raman: Have started working on Approach B for the newer approach utilising RoBERTa and weighted class loss to heavily penalize the errors on minority classes.
- Notebooks are in their nascent stage, full pipeline is not yet ready.

Results and Analysis (As per current status)

1. **Current Status:**

Data readiness: Dataset (Reddit Mental Health) identified and mapped to 3-tier severity levels.

Model setup: Environment set up for both Approach A and Approach B.

2. Defined evaluation metrics:

Primary metric: Macro-F1 score

Critical metric: Recall for "Severe" class, since a False Negative is more dangerous than False Positive.

3. Comparative Analysis: Upon initial analysis we are expecting that Approach B will be better than Approach A on Reddit data because Approach B is optimized for informal text/slang which is typically the mode of communication in informal conversations between patients and their counsellors.

Individual Contribution of each student

Sl.no	Name – Reg. no	Details
1	Pranav Bhat (231IT049)	Implementation of Approach B (RoBERTa + Custom Loss Function).
2	Rijul Raman (231IT056)	Literature review, Dataset acquisition, and cleaning (Regex/Preprocessing).
3	Pranav Vinodh (231IT050)	Implementation of Approach A (BERT + Augmentation Pipeline)
4	Sanchit Srisha Udupa (231IT060)	Performance Evaluation, Metric calculation (Precision/Recall/F1), and Comparative Analysis.

References

- [1] A. Abilkaiyrkyzy et al., "Dialogue System for Early Mental Illness Detection: Toward a Digital Twin Solution," *IEEE Access*, vol. 12, 2024.
- [2] Mental Health Text Classification Dataset (Reddit), Hugging Face.
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- [4] Y. Liu et al., "RoBERTa: A Robustly Optimized BERT Pretraining Approach," *arXiv preprint arXiv:1907.11692*, 2019.\
- [5] A. Haque et al., "Deep Learning for Suicide and Depression Identification with Unsupervised Label Correction," in *Proc. ICANN*, 2021.